

AI-Powered FRA Atlas and WebGIS Decision Support System for Forest Rights Management

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1. Problem Statement

Under the Forest Rights Act (FRA) 2006, tribal communities and forest dwellers are granted legal rights over forest land. However, the current system faces several challenges:

- Land records are stored in paper documents.
- Information is scattered across different government offices.
- Verification of claims is time-consuming.
- Officers cannot easily visualize land locations.
- There is no centralized platform for monitoring FRA beneficiaries.
- Difficult to identify eligible government schemes for beneficiaries.
- Lack of data-driven decision making.

As a result, processing claims, monitoring land usage, and providing benefits become inefficient.

2. Proposed Solution

We propose an AI-Powered FRA Atlas and WebGIS Decision Support System that:

Digitizes FRA Records

Upload scanned documents

Extract information using OCR

Store records in a centralized database

GIS Mapping

Display all FRA lands on an interactive map

Visualize villages, forests, water bodies, and beneficiary locations

Satellite Image Analysis

Analyze land cover using AI

Identify forest areas, agriculture, water bodies, and settlements

Decision Support Engine

Recommend suitable government schemes

Help officers prioritize beneficiaries

Analytics Dashboard

Real-time statistics

Claim approval monitoring

Land distribution analysis

3. Why Our Solution is Better Than Existing Systems

Paper-based records → Fully digital records

Manual verification → AI-assisted verification

No map visualization → Interactive GIS maps

Separate databases → Centralized platform

No recommendations → Intelligent scheme suggestions

Difficult reporting → One-click analytics and reports

Time-consuming search → Instant search and filtering

Key Innovation

Most systems only store records.

Our system:
Stores records
Maps records
Analyzes records
Generates recommendations

This makes it a Decision Support System, not just a management system.

4. Technology Stack

Frontend
React.js
Vite
Tailwind CSS
Leaflet.js
Recharts

Backend
FastAPI
SQLAlchemy
JWT Authentication

AI/ML
Tesseract OCR
spaCy NLP
TensorFlow

GIS Technologies
Leaflet.js
OpenStreetMap
PostGIS

Deployment
Docker
Docker Compose

5. Database Used

Primary Database
PostgreSQL

Stores:
Users
Beneficiaries
Documents
Reports
Recommendations

Why PostgreSQL?
Open source
Highly scalable
Secure
Supports geospatial data

GIS Database Extension
PostGIS

Stores:

Latitude

Longitude

Land Boundaries

Village Boundaries

Forest Polygons

Used for:

Map visualization

Geospatial queries

Distance calculations

6. System Workflow

FRA Documents → OCR Processing → Data Extraction → PostgreSQL Database → GIS Mapping
→ Satellite Analysis → Decision Support Engine → Dashboard & Reports

7. System Architecture

FRONTEND (React + Tailwind + Leaflet + Recharts)

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FASTAPI BACKEND

Authentication Module

Beneficiary Management Module

OCR Module

GIS Module

Recommendation Engine

Report Generator

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AI LAYER

Tesseract OCR

spaCy NLP

TensorFlow Satellite Analysis

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DATABASE LAYER

PostgreSQL

PostGIS

8. Detailed Data Flow Diagram

Government Officer → React Web Portal → FastAPI Backend → OCR Module / GIS Module / AI Engine → PostgreSQL + PostGIS → Dashboard & Reports

9. Expected Outcomes

70–80% reduction in manual record processing.

Faster beneficiary verification.

Better land monitoring through GIS.

Improved scheme allocation.

Centralized digital record management.

Data-driven government decision making.

10. One-Line Project Description (for Resume)

"Developed an AI-powered WebGIS platform that digitizes Forest Rights Act records, visualizes land ownership on interactive maps, analyzes satellite imagery, and provides intelligent decision support for government officers using React, FastAPI, PostgreSQL, PostGIS, OCR, and Machine Learning."